

Lipkin model, Entanglement, Noise, Dephasing and dissipation

Juan David Álvarez-Cuartas

*Department of Physics and Centre for Bioinformatics and Photonics—CIBioFi,
Universidad del Valle, Cali 760032, Colombia and*

Department of Physics and Center for Computing in Science Education, University of Oslo, N-0316 Oslo, Norway

Patrick Cook and Danny Jammooa

*Facility for Rare Isotope Beams and Department of Physics and Astronomy,
Michigan State University, East Lansing, Michigan 48824, USA*

Morten Hjorth-Jensen

Department of Physics and Center for Computing in Science Education, University of Oslo, N-0316 Oslo, Norway

Dean Lee

*Facility for Rare Isotope Beams and Department of Physics and Astronomy,
Michigan State University, East Lansing, Michigan 48824, USA*

Juan M. Scarpetta, John H. Reina

*Department of Physics and Centre for Bioinformatics and Photonics—CIBioFi,
Universidad del Valle, Cali 760032, Colombia and*

Department of Physics and Center for Computing in Science Education, University of Oslo, N-0316 Oslo, Norway

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I. INTRODUCTION

Structure of article

1. Introduction with motivation
2. Discussion of the Lipkin model
3. Setting up time evolution
4. PMMs and time evolution
- 5.

II. MATHEMATICAL FRAMEWORK AND ENTANGLEMENT MEASURES

A. Hamiltonian and Symmetry.

The Lipkin–Meshkov–Glick (LMG) model describes N spin- $\frac{1}{2}$ particles interacting through collective spin operators. Its Hamiltonian can be expressed as

$$\hat{H}_{\text{LMG}} = -\frac{\lambda}{N} \left(\hat{S}_x^2 + \gamma \hat{S}_y^2 \right) - h \hat{S}_z, \quad (1)$$

where $\hat{S}_\alpha = \frac{1}{2} \sum_{i=1}^N \hat{\sigma}_i^\alpha$ are collective spin operators, λ is the interaction strength, γ the anisotropy parameter, and h an external magnetic field. The factor $1/N$ ensures a well-defined thermodynamic limit. The total spin $\hat{S}^2$ commutes with the Hamiltonian, implying conservation of total spin $S = N/2$, and the system therefore lives in the symmetric subspace spanned by the Dicke states

$$|S, M\rangle \equiv |N/2, M\rangle, \quad M = -S, \dots, S. \quad (2)$$

B. Mean-Field and Quantum Phase Transition.

In the thermodynamic limit ($N \rightarrow \infty$), a mean-field treatment becomes exact. A semiclassical energy functional can be written in terms of collective spin variables (θ, ϕ) on the Bloch sphere as

$$E(\theta, \phi) = -\frac{\lambda}{2} \sin^2 \theta (\cos^2 \phi + \gamma \sin^2 \phi) - h \cos \theta. \quad (3)$$

The ground state undergoes a second-order quantum phase transition at $h_c = \lambda(1 + \gamma)/2$, separating a symmetric phase ($\langle S_x \rangle = 0$) from a symmetry-broken phase ($\langle S_x \rangle \neq 0$). Near this critical point, the entanglement entropy exhibits logarithmic scaling with system size, signalling a diverging correlation length in the collective degrees of freedom.

C. Reduced Density Matrices and Entropy.

Because of full permutation symmetry, the reduced density matrix of any single spin is identical and can be obtained from the collective spin density operator ρ as

$$\rho_1 = \text{Tr}_{2,\dots,N}(\rho) = \frac{1}{2} \left(\mathbb{I} + \frac{2}{N} \langle \hat{S} \rangle \cdot \boldsymbol{\sigma} \right). \quad (4)$$

The single-spin entanglement (or *one-tangle*) with the remainder of the system is quantified by the linear entropy

$$\tau_1 = 4 \det(\rho_1) = 1 - \frac{4}{N^2} \langle \hat{S} \rangle^2. \quad (5)$$

This quantity vanishes in product states and reaches unity for maximally mixed single-spin states.

For bipartite entanglement between two subsystems of sizes L and $N - L$, the von Neumann entropy

$$S_L = -\text{Tr}(\rho_L \log_2 \rho_L) \quad (6)$$

is the standard measure. In the symmetric Dicke basis, ρ_L is block-diagonal and its eigenvalues can be expressed analytically in terms of Clebsch–Gordan coefficients, allowing explicit evaluation of S_L for moderate N .

D. Two-Spin Correlations and Concurrence.

Pairwise entanglement between two spins is characterized by the *concurrence* C , defined from the two-spin reduced density matrix ρ_{12} as

$$C = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}, \quad (7)$$

where λ_i are the square roots of the eigenvalues (in descending order) of the matrix $\rho_{12}\tilde{\rho}_{12}$, with $\tilde{\rho}_{12} = (\sigma_y \otimes \sigma_y) \rho_{12}^* (\sigma_y \otimes \sigma_y)$. In the LMG model, due to full permutation symmetry, C is identical for all spin pairs. It reaches a maximum near the critical point and decays as $N^{-1/3}$ at criticality, indicating that bipartite entanglement is sub-extensive even though the total multipartite entanglement diverges.

E. Multipartite and Mixed-State Measures.

Beyond bipartite partitions, the *global entanglement* and the *geometric measure of entanglement* are often used:

$$E_G = 1 - \max_{|\phi_{\text{sep}}\rangle} |\langle \phi_{\text{sep}} | \psi_0 \rangle|^2, \quad (8)$$

where $|\phi_{\text{sep}}\rangle$ is the closest separable state to the ground state $|\psi_0\rangle$. For mixed or non-complementary partitions, the logarithmic negativity

$$\mathcal{N} = \log_2 \|\rho^{TA}\|_1 \quad (9)$$

quantifies entanglement by means of the trace norm of the partially transposed density matrix ρ^{TA} .

F. Scaling and Universality.

Near the quantum critical point, different entanglement measures show universal scaling with system size N :

$$S_L \sim \frac{1}{6} \log_2 N + \text{const}, \quad \tau_1 \sim N^{-2/3}, \quad C \sim N^{-1/3}. \quad (10)$$

These exponents coincide with those predicted by conformal field theory for one-dimensional critical systems, even though the LMG model is infinite-range. Such universality underscores that entanglement, rather than spatial locality, captures the essential signatures of quantum criticality.

III. RESULTS AND DISCUSSIONS

IV. CONCLUSION